

Problem set 3: Market Failure

Market Failure

1. Specify three economic reasons for a market failure. Give one example of each reason and explain how legal intervention can solve the underlying problem.
2. Is the following statement correct? “*Monopolies are inefficient because they charge a price that is too high*”. If not, explain why monopolies may be inefficient.
3. Explain why it may be difficult to adopt a global policy to fight off climate change, based on the problem of public goods.
4. A bank manager wants to increase the interest rate on consumer loans. One of their advisors say that this is not a good idea because of “adverse selection”. Explain what the advisor means by this.
5. James and Joe live across the street from one another. They would be happy to exchange houses, but every time one of them makes an offer – the other refuses, saying “my house is worth more”. Explain why such a market failure might occur based on a behavioral reason.

Market intervention

6. Answer the following sections (a-c):
 - a. In a market with standard supply and demand curves, the government decides to impose a 2\$ tax on every produced unit. Draw a diagram of the supply and demand curves before and after the tax is imposed. Mark the area in which there is a deadweight loss, if any.
 - b. Now suppose instead that the product is suddenly perceived as crucial for living and, as a result, the demand becomes perfectly inelastic. Explain who will bear more of the tax as a result (the supplier or the consumers) and why.
 - c. Next, suppose instead that the supplier is a monopoly. Will this tax then exacerbate the inefficiency due to the market power or mitigate it?
7. Which of the following statements is correct? (mark all the correct answers)
 - a. Imposing a per-unit tax causes a deadweight loss.
 - b. A per-unit tax entails a deadweight loss, but a per-unit subsidy does not.
 - c. Imposing a lump sum tax does not cause a deadweight loss.
 - d. Taxing a monopoly with a per-unit tax will cause it to produce more.
 - e. A high lump sum tax might discourage firms from entering the market

8. Suppose that the state of Hamburg imposes a fixed “Mietbremse”, where no landlord is allowed to charge more than 1500 EUR for an apartment. What will be the (possible) effects of this decision?
9. Explain why maximizing the total surplus in the market for a certain product does not, necessarily, guarantee the maximization of social welfare.
10.
 - a. State an economic situation with hidden information. Explain why hidden information can lead to adverse selection? What is the consequence for market efficiency?
 - b. Explain for the situation you described in (a) how signaling can improve the market outcome and why it allows the “good type” to distinguish from the “bad type”.